



Wrangle Analyze Data

Report for Udacity Data Analysis Nanodegree, **Wrangle Analyze Data** project.

Wrangle Report

The dataset wrangle in the project is the twitter archive of the Twitter user **@dog_rates** ("WeRateDogs 🐶")

There are three steps of data wrangling process:

- ✓ Gathering Data
- ✓ Assessing Data
- ✓ Cleaning Data

And some goals of this project:

- ✓ Storing, analyzing & visualizing wrangled data
- ✓ Reporting

Gathering Data

I have gathered data from the following sources:

- `twitter_archive_enhanced.csv` file that provided from Udacity to us. It contains basic data about tweets (tweet id, timestamp and etc.).

```
# Columns of this dataset
[
  'tweet_id', 'in_reply_to_status_id', 'in_reply_to_user_id', 'timestamp',
  'source', 'text', 'retweeted_status_id', 'retweeted_status_user_id',
  'retweeted_status_timestamp', 'expanded_urls', 'rating_numerator',
  'rating_denominator', 'name', 'doggo', 'floofer', 'pupper', 'puppo'
]
```

- `image_predictions.tsv` contains image predictions and it also provided by Udacity. We should download these predictions file by using python's request library and by using this library we can know information about breed of dog (this predictions are made by neural networks).
- I had to use `tweepy` python library for gathering additional useful information about each tweet which is presented in the Twitter archive file. After making requests, I have retweets count, likes count and some other interesting data that I found.

Assessing Data

After gathering data, I started to assess data both visual and programmatic.

Quality Issues:

- `twitter_archive_enhanced.csv`
 - Missing data in the some columns.
 - Some dog names are unusual (*a,aa an ...*)
 - Timestamp column is an `object`
 - Rating denominator is not standardized.
 - `tweet_id`s should be a string.
- `image_predictions.tsv`
 - Duplicates on `jpg_url`

● `tweet_json.txt`

- Missing some data.

Tidiness Issues:

`twitter_archive_enhanced.csv`

- Timestamp should be datetime.
- None values in the dog names, I think there should be `NaN` (`np.nan`)
- There should be one column for the various dog types: *doggo*, *floofer*, *pupper*, *puppo*.
- Missing `dog_breed` column, it will be useful for analysis process.

`image_predictions.tsv`

- Image prediction table should be with twitter archive table.

`tweet_json.txt`

- This also should be with twitter archive table.

Cleaning Data

After assessment process, I cleaned the dataset.

Define  Code  Test:

- ✓ Merged `image_predictions` and `tweet_json` to twitter archive
- ✓ Created two new columns: date and time columns by using timestamp column
- ✓ Dropped duplicated `jpg_url`s from image prediction
- ✓ Dropped dog names which are less than 3 chars
- ✓ I fixed and standardized dog's ratings & created new one `rating` column
- ✓ Converted `None` values in the dog names to `np.nan`
- ✓ Created one column for the various dog types (doggo, floofer, pupper & puppo)
- ✓ Created new `dog_breed` column using image predictions data